

Reviewer Pack — Minimal Rank of Twisted Tensor Products vs Monotone Circuit ...

Entry a05be7217792 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Twisted Tensor Products vs Monotone Circuit Size for k-CNF

Entry ID: a05be7217792 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 03:07:41 UTC

1. Conjecture

Field A (mathematical branch): Twisted Tensor Product Theory **Field B** (complexity object): Complexity Theory: Monotone Circuit Complexity

Statement:

[‘For a given n , there exists a constant c such that the minimal rank of the twisted tensor product of two random n -dimensional vectors is at least cn .’, ‘This inequality holds for all instances with $n \leq 40$ and property P holds if for every instance in this size range, property Q holds.’, ‘Property P : The twisted tensor product of the vectors is a permutation matrix.’]

Rationale (proposer’s reasoning):

[‘Twisted tensor products are a generalization of tensor products that have not been extensively applied to complexity theory. Their algebraic structure and symmetry properties could provide insights into circuit complexity problems.’, ‘Monotone circuits are a restricted class of Boolean circuits, whose lower bounds are important for understanding the limits of computation. If twisted tensor products can be used to bound monotone circuits, it might lead to new proof techniques in complexity theory.’]

Taxonomy category: GCT_DET_PERM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e3a16c1c7337ce4a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all $n \leq 40$, the average minimal rank of the twisted tensor product across 30 seeds is at least cn with a p -value ≤ 0.05 , where ‘ cn ’ is defined as the expected value under the null hypothesis and c is the constant from the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (1): - "minimal rank" twisted tensor product AND monotone circuit complexity", "permutation matrix" IN Twisted Tensor Product Theory AND k-CNF", "property P" AND "property Q" in Complexity Theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 40
    vectors = [random.choices(range(-10, 11), k=n) for _ in range(2)]
    tensor_product = [[sum(a * b for a, b in zip(v1, v2)) for v2 in vectors[1]] for v1 in vectors[0]]

    def is_permutation_matrix(matrix):
        n = len(matrix)
        if any(len(row) != n for row in matrix): return False
        identity = [[1 if i == j else 0 for j in range(n)] for i in range(n)]
        for i in range(n):
            if sum(matrix[i]) != 1 or sum(matrix[j][i] for j in range(n)) != 1: return False
        return True

    min_rank = float('inf')
    for row in tensor_product:
        rank = len(set(row))
        if rank < min_rank:
            min_rank = rank

    conjecture_holds = is_permutation_matrix(tensor_product)
    counterexample = "Non-permutation matrix" if not conjecture_holds else ""

    return {
        "metric_name": "min_rank",
        "metric_value": min_rank,
```

```

        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2**i + 3 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efb317c7.py", line 56, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_efb317c7.py", line 23, in run_trial
    tensor_product = [[sum(a * b for a, b in zip(v1, v2)) for v2 in vectors[1]] for v1 in vectors[0]]
                      ^^^^^^^^^^^^^
TypeError: 'int' object is not iterable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating whether the conjecture is supported or falsified according to the pre-registered support condition.
| next: Investigate and fix the error in the test code to allow for a proper evaluation of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11988
2	propose	ollama_remote	glm4:latest	0	0	10921
3	propose	ollama_remote	glm4:latest	0	0	13422
4	preregistration	ollama_remote	glm4:latest	0	0	5532
5	novelty	ollama_remote	glm4:latest	0	0	4625
6	novelty	ollama_remote	glm4:latest	0	0	5186
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	34050
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8455
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8539
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7066
11	judge	ollama_remote	glm4:latest	0	0	10001

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 119785 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in *research/audit/a05be7217792.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/a05be7217792.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/a05be7217792.tar.gz* (if generated)